Course Title	Data Science and AI Project (SPRING)Cap.81		
Class Code	B3E6020101		
Instructor	DUBEY Priya		
Semester / Year	2026年度 春学期	Campus	Ikebukuro Campus
Timetable	Mon.4Period,Thu.4Period	Credits	4Credits
Enrollment Semester	semester not specified	Course Leve	M-INF320
Department / School	EC		
Course Description	This course is an undergraduate major course that is highly recommended for students pursuing a specialization in Digital Business and Innovation and IT Business. The course offers an in-depth, hands-on introduction to Data Science and Artificial Intelligence, using Python as the core programming language. Starting with essential programming foundations, the course progresses through key topics such as data analysis, data visualization, machine learning, and model evaluation. Libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn will be used extensively. Emphasis will be placed on practical implementation through coding, real-world datasets, and problem-solving.		
Objectives	Students will be able to use Python and Jupyter for data science tasks. Students will be able to apply data cleaning, EDA, and visualization techniques. Students will be able to understand and implement supervised and unsupervised machine learning algorithms. Students will be able to evaluate models using appropriate metrics. Students will be able to plan, execute, and present a real-world AI/Data Science capstone project.		
Recommended Courses	[Prerequisite] Foundations of Python [Recommended (but not prerequisite)] Database and Big Data, AI & Machine Learning for Decision Making. Text Mining and Deep Learning [Others] None		
Relevance to D.P.	Please refer to " here" to see how this course is related to your School/Department Diploma Policy(DP).		
Coursework Time	At TIU, 1 credit corresponds to 45 hours of coursework time, which includes not only the time a student spends in the classroom but also the hours of study in his or her own time, such as approximately 4 hours per week of outside class study for lectures/seminars and 1 hour for Foreign Language courses respectively . Please refer to " here" for more details. For details on outside class study, please check the below "Outside Class Study (Preparation / Review)" .		

Textbooks			
Required textbooks	None	Additional Details	Class Materials will be uploaded on Moodle or distributed during the class.
Title	N/A		
Author	N/A		
Publisher	N/A		
Year published	N/A		
ISBN	N/A		
Price	N/A		
Reference Books			

Method of Inst					
Group Work	○	Presentation	○	Discussion, Debate	○
Project Based Learning	○	Flip Teaching	○	Practice, Field Work	○
Additional Details					

Utilization of Moodle					
Quiz	○	Assignment	○	Forum	○

Feedback	○	Survey	○	Choice	○
Additional Details					

Course Outline		Outside Class Study (Preparation/Review)
Class1	Lecture: Course Introduction- Data Science & AI in Practice	【Preparation】Not Required 【Review】Download and install Anaconda or set up Google Colab for future programming. Explore the interface.
Class2	Lecture and group programming: Python Essentials for Data Science	【Preparation】Programming lectures of Foundations of Python 【Review】Practice basic Python syntax. Write small programs using loops and conditionals.
Class3	Lecture: Jupyter Notebooks, IDEs, and Package Management	【Preparation】Not Required 【Review】Practice installing packages (e.g., numpy, pandas). Create and run simple notebooks.
Class4	Lecture and group programming: Data Structures in Python	【Preparation】Not Required 【Review】Re-write and modify class code. Practice converting between different data structures.
Class5	Lecture and group programming: Numpy for Numerical Computing	【Preparation】Not Required 【Review】Practice NumPy indexing, reshaping, broadcasting. Try numerical operations and summaries.
Class6	Lecture and group programming: Pandas for DataFrames and Tabular Data	【Preparation】Not Required 【Review】Practice loading data with read_csv(), data selection, filtering, and aggregation.
Class7	Lecture and group programming: Data Cleaning: Handling Missing and Duplicate Data	【Preparation】Not Required 【Review】Apply cleaning techniques to a messy dataset. Try out data imputation methods.
Class8	Lecture and group programming: Exploratory Data Analysis (EDA)	【Preparation】Not Required 【Review】Practice visual and statistical summaries. Write Python functions to summarize data features.
Class9	Lecture and group programming: Data Visualization with Matplotlib & Seaborn	【Preparation】Not Required 【Review】Create plots: histograms, bar plots, scatter plots.
Class10	Lecture and group programming: Correlation, Outliers, and Distributions	【Preparation】Not Required 【Review】Review the last lecture, Visualize outliers using boxplots or scatter plots.
Class11	Lecture and group programming: Introduction to Machine Learning	【Preparation】Not Required 【Review】Review code from class. Write simple classification or regression model using scikit-learn.
Class12	Lecture and group programming: Supervised Learning: Classification	【Preparation】Not Required 【Review】Train classifiers on small datasets. Evaluate performance using accuracy and confusion matrix.
Class13	Lecture and group programming: Logistic Regression and Decision Trees	【Preparation】Not Required 【Review】Train both models on a dataset. Compare their predictions.
Class14	Lecture and group programming: Model Evaluation: Accuracy, Precision, Recall	【Preparation】Not Required 【Review】Review the last lecture, Compute and interpret these metrics for different models.
Class15	Lecture and group programming: Cross Validation and Hyperparameter Tuning	【Preparation】Not Required 【Review】Apply GridSearchCV on an ML model. Tune parameters like depth, learning rate.
Class16	Lecture and group programming: Regression Models: Linear, Ridge, Lasso	【Preparation】Not Required 【Review】Review the last lecture, practice all the codes.
Class17	Lecture and group programming: Feature Engineering and Selection	【Preparation】Not Required 【Review】Review the last lecture, practice all the codes.

Class18	Lecture and group programming: Introduction to Unsupervised Learning	【Preparation】Not Required 【Review】Practice fitting unsupervised models like KMeans. Analyze clustering results.
Class19	Lecture and group programming: Clustering: KMeans and Hierarchical	【Preparation】Not Required 【Review】Review the last lecture, re-write all the codes again and Visualize clusters using scatter plots.
Class20	Lecture and group programming: Dimensionality Reduction Techniques	【Preparation】Not Required 【Review】Review the last lecture, apply PCA and Interpret variance.
Class21	Lecture and group programming: Introduction to Capstone: Use Cases + Dataset Selection	【Preparation】Not Required 【Review】Choose a dataset and define the problem statement. Start initial EDA.
Class22	Lecture and group programming: Project Phase 1: Problem Framing & Data Analysis	【Preparation】Not Required 【Review】Refine EDA based on peer/instructor feedback. Document findings and assumptions.
Class23	Lecture and group programming: Modeling and Evaluation	【Preparation】Not Required 【Review】Evaluate different machine learning models and track performance. Create charts to compare models.
Class24	Lecture and group programming: Case Study: Fraud Detection	【Preparation】Not Required 【Review】Apply learned techniques on a real or synthetic fraud dataset.
Class25	Lecture and group programming: Case Study: Air BnB data analytics	【Preparation】Not Required 【Review】Perform EDA and visualization on Air bNb dataset and summarize your findings.
Class26	Lecture and group programming: Presentation and Report	【Preparation】Not Required 【Review】Finalize your project report and findings.

Evaluation Components

Class Evaluation	20	Details	Your grade for “Class Evaluation” is not only defined by attending the classes, but also your active participation, optional presentations, Assignment submission (including Moodle submissions) for each class meeting
Examination	50	Details	The examination for this course consists of two assignments and a final quiz. Assignment 1 (15 marks) and Assignment 2 (15 marks) assess students’ understanding of fundamental data science concepts and applied artificial intelligence techniques through written and analytical tasks. The final quiz (20 marks) evaluates overall understanding of the key concepts covered in the course.
Report	30	Details	Students must submit a project report that clearly describes a data science or artificial intelligence project. The report should include a problem statement, description of the dataset, methodology, model selection, evaluation approach, results interpretation, and limitations. The report should demonstrate logical reasoning, clarity of explanation, and appropriate application of course concepts.
Others	0	Details	

Assignment Feedback	There will be assignments and feedback on Moodle for each task
Professor Contact Hours	Students are encouraged to raise any questions at the conclusion of each class session in order to clarify the material presented. For inquiries that require additional discussion, you may contact me via email at pdubey@tiu.ac.jp
Remarks	To enforce the TIU Student Code of Conduct, the instructor in this course may deduct up to 20% off a student’s total course score if a student does not comply with direct instructions given on class conduct by the instructor.